

DELTA COUNTY AP, MI, USA

WMO: 726480

Lat: 45.733N

Lon: 87.083W

Elev: 594

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-3.7	-18.4	2.1	-4.4	-14.2	2.6	-2.4	23.6	18.5	20.6	18.1	6.2	270	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	82.2	69.5	80.6	68.7	77.3	67.2	73.1	79.5	71.0	77.2	69.1	75.0	11.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	116.9	77.8	69.5	110.8	76.0	66.4	99.3	73.1	37.0	80.3	35.2	77.5	33.6	75.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.6	16.8	DB	-15.2	88.5	5.9	3.3	-19.5	90.9	-23.0	92.8	-26.3	94.7	-30.6	97.1
			WB	-16.4	75.9	5.9	2.3	-20.6	77.5	-24.0	78.9	-27.3	80.2	-31.6	81.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	17.4	18.6	27.7	38.9	50.8	60.9	65.9	64.8	57.9	46.2	34.8	24.2	(6)
	DBStd	19.03	10.80	11.05	10.03	7.18	7.19	6.69	5.93	5.53	7.20	7.65	8.51	9.68	(7)
	HDD50	4445	1010	878	692	338	78	4	0	0	14	170	460	801	(8)
	HDD65	8436	1475	1298	1156	783	443	158	62	72	232	584	907	1266	(9)
	CDD50	1696	0	0	1	4	103	330	494	458	251	52	3	0	(10)
	CDD65	213	0	0	0	0	3	35	91	65	18	1	0	0	(11)
	CDH74	1413	0	0	0	2	32	240	678	385	74	2	0	0	(12)
	CDH80	230	0	0	0	0	6	43	134	42	5	0	0	0	(13)
	Wind	WSAvg	7.8	8.3	8.1	8.0	8.4	7.9	7.0	6.9	7.5	8.2	8.5	8.2	(14)
	Precipitation	PrecAvg	30.2	1.4	1.0	1.9	2.5	3.3	3.3	3.1	3.5	3.0	2.2	1.5	(15)
(16) Precipitation	PrecMax	37.7	3.8	1.9	4.2	5.4	6.2	7.2	8.1	4.5	8.0	5.5	4.6	2.9	(16)
	PrecMin	24.4	0.4	0.0	0.4	0.7	1.3	1.4	0.8	1.0	1.3	0.9	0.8	0.2	(17)
	PrecStd	3.7	0.8	0.5	1.1	1.1	1.4	1.3	1.6	0.9	1.5	1.5	1.0	0.7	(18)
	PrecStd	3.7	0.8	0.5	1.1	1.1	1.4	1.3	1.6	0.9	1.5	1.5	1.0	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	44.0	56.6	64.4	78.9	84.4	88.1	83.9	80.6	70.4	58.9	47.6	(19)
		MCWB	37.4	38.8	47.5	50.6	61.8	67.8	71.3	71.2	68.5	62.9	51.5	45.6	(20)
	2%	DB	37.2	39.1	48.4	57.5	72.1	80.6	83.5	81.1	75.4	65.7	54.0	43.1	(21)
		MCWB	34.5	35.4	41.9	46.6	59.2	67.1	70.9	69.6	66.3	60.3	49.3	39.9	(22)
	5%	DB	35.7	36.4	45.1	54.7	67.5	76.3	81.1	78.8	73.0	63.0	50.8	39.2	(23)
		MCWB	33.8	33.5	39.2	44.9	56.9	65.6	69.6	68.2	65.6	57.0	46.8	36.4	(24)
	10%	DB	33.4	34.2	41.0	51.9	63.7	72.9	78.5	75.3	71.0	59.6	47.7	36.8	(25)
		MCWB	31.8	31.6	35.9	43.2	54.6	62.9	68.0	66.5	64.5	54.7	44.4	34.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	40.3	49.7	54.2	65.9	73.1	76.2	74.4	72.0	65.4	53.4	46.0	(27)
		MCDB	40.8	43.0	55.4	60.9	72.9	81.6	83.1	80.7	77.6	67.6	57.8	46.8	(28)
	2%	WB	35.4	36.3	43.5	49.6	61.7	69.3	73.8	72.4	69.2	61.5	50.1	40.6	(29)
		MCDB	37.0	38.8	47.3	56.1	68.8	77.2	80.2	78.5	73.3	64.3	53.1	43.1	(30)
	5%	WB	33.3	33.6	39.6	46.3	58.7	67.0	71.5	70.3	67.3	57.9	46.9	37.6	(31)
		MCDB	34.9	36.0	44.5	52.8	65.8	73.9	77.9	76.0	71.7	62.2	49.2	39.5	(32)
	10%	WB	31.1	31.5	36.8	43.7	55.5	64.9	69.5	68.4	65.3	55.1	44.4	34.7	(33)
		MCDB	32.7	33.9	40.5	50.6	62.7	71.1	75.7	73.9	69.5	58.9	47.2	36.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	17.0	16.6	17.4	19.3	19.0	19.8	19.0	18.5	16.4	13.4	13.2	(35)
		MCDBR	14.1	15.3	18.3	23.7	25.3	22.1	22.0	20.4	20.2	19.7	16.7	13.8	(36)
		MCWBR	12.5	12.5	13.2	15.2	14.6	12.4	12.4	12.0	13.0	14.3	13.2	12.2	(37)
	5% WB	MCDBR	13.2	13.8	16.9	21.7	21.3	18.8	19.9	17.9	17.2	17.0	15.1	13.0	(38)
		MCWBR	12.1	11.6	12.8	15.2	14.2	11.8	12.6	11.9	12.0	13.8	13.1	12.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.258	0.276	0.315	0.349	0.384	0.385	0.383	0.380	0.360	0.328	0.298	0.267	(40)
	taud		2.469	2.389	2.386	2.386	2.352	2.390	2.413	2.415	2.465	2.538	2.540	2.496	(41)
	Ebn at Noon		276	289	291	289	280	280	279	276	272	268	258	260	(42)
	Edh at Noon		21	28	32	36	38	37	36	34	30	24	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		407	708	1117	1453	1775	1911	1964	1662	1259	758	426	314	(44)
	RadStd		36	85	103	138	138	151	85	110	91	83	44	29	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page